SECURITY EAGLES

eJPTv2

Junior Penetration Tester Certification

STUDY GUIDE

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FROM LEARNERS TO LEADERS

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Chapter 1

Getting Started

1.1 Introduction to Penetration Testing

Penetration testing, often called **pen testing** or **ethical hacking**, is a simulated cyber-attack against a computer system, network, or web application to identify security vulnerabilities that an attacker could exploit. Unlike malicious hackers, penetration testers have explicit permission from the organization to probe their systems and report findings.

1.1.1 What is Penetration Testing?

Key Objectives

- Identify security weaknesses before attackers do
- Test the effectiveness of security controls
- Validate compliance with security policies
- Provide actionable recommendations for improvement

1.1.2 Types of Penetration Tests

1. Black Box Testing

- Tester has no prior knowledge of the target system
- Simulates an external attacker's perspective
- Most time-consuming but realistic approach

2. White Box Testing

- Tester has complete knowledge of the infrastructure
- Access to source code, architecture diagrams, credentials
- More thorough but less realistic of external threats

3. Grey Box Testing

- Tester has partial knowledge (e.g., user-level access)
- Balances realism with efficiency
- Most common in real-world engagements

1.1.3 Penetration Testing Methodology

The penetration testing process follows a structured approach:

1. **Planning and Reconnaissance**
   Gathering information about the target, defining scope, and understanding the environment.
2. **Scanning and Enumeration**
   Identifying live hosts, open ports, services, and potential entry points.
3. **Vulnerability Assessment**
   Analyzing discovered services for known vulnerabilities and misconfigurations.
4. **Exploitation**
   Attempting to exploit identified vulnerabilities to gain unauthorized access.
5. **Post-Exploitation**
   Maintaining access, privilege escalation, lateral movement, and data exfiltration.
6. **Reporting**
   Documenting findings, risk ratings, and remediation recommendations.

1.1.4 Legal and Ethical Considerations

Critical Rules

- Always obtain **written authorization** before testing any system
- Stay within the defined scope of engagement
- Avoid causing damage or disrupting services
- Maintain confidentiality of discovered information
- Follow responsible disclosure practices

1.1.5 Rules of Engagement (ROE)

Before any penetration test, establish clear rules:

- **Scope:** What systems, networks, or applications are in scope?
- **Timeline:** When can testing occur? (avoid business-critical hours)

- **Methods:** What techniques are allowed? (DoS attacks usually excluded)
- **Communication:** Who to contact in case of critical findings?
- **Data handling:** How to handle sensitive data discovered during testing?

1.2 eJPT Certification Overview

1.2.1 What is eJPT?

The eLearnSecurity Junior Penetration Tester (eJPT) is an entry-level certification designed to validate fundamental penetration testing skills. It's offered by INE (formerly eLearnSecurity) and is recognized as an excellent starting point for aspiring penetration testers.

1.2.2 Why eJPT?

- Practical, hands-on exam (no multiple choice)
- Affordable compared to other certifications
- Industry-recognized credential
- Beginner-friendly entry to penetration testing
- Builds a strong foundation for advanced certifications

1.2.3 Exam Format and Structure

Exam Details

- **Duration:** 48 hours (2 days)
- **Format:** 100% practical, hands-on lab exam
- **Environment:** Realistic network with multiple vulnerable targets
- **Passing Score:** 70% required to pass
- **Attempts:** 2 attempts
- **Tools:** The exam provides an in-browser, pre-configured Kali Linux environment with the tools, scripts and wordlists you need. The Kali instance has no internet access, but you may use your host machine's browser for research.

1.3 Basic Networking Fundamentals

1.3.1 What Is a Network?

A network is a group of devices (computers, phones, servers, routers) that can talk to each other and share information. Every penetration test happens on a network, so understanding the basics is essential.

Simple Example

Your laptop → connected to Wi-Fi → connected to the Internet.
That's a network.

1.3.2 IP Addresses (Your Device's Identity)

An IP address is like a home address for your device on a network.

Example

`192.168.1.2`

Two important parts:

- **Network ID** → identifies the network
- **Host ID** → identifies the device inside that network

Private IP Ranges (most common in labs)

- `10.x.x.x`
- `172.16.x.x - 172.31.x.x`
- `192.168.x.x`

When hacking inside a company network or lab, you will almost always see these.

1.3.3 Subnet Mask (Network Size)

A subnet mask tells you how big your network is.

Example

`255.255.255.0` → `/24`

What you should remember:

- `/24` networks are used most in eJPT labs
- They have 254 usable devices
- Example network: `192.168.1.0/24`
- Broadcast address: `192.168.1.255`

Note: You don't need deep subnetting for eJPT — just understand /24.

1.3.4 MAC Address

A MAC address is the physical ID of your network card.

Example

```
A4:12:BC:93:1F:20
```

It never changes and is used inside local networks (like inside your home).

1.3.5 The OSI Model

The OSI (Open Systems Interconnection) model breaks networking down into 7 layers. As a pentester, you focus heavily on **Layers 2, 3, 4, and 7**.

1. **Layer 1 (Physical):** Cables and Wi-Fi signals.
2. **Layer 2 (Data Link):** Uses MAC Addresses to move data between devices on the same network.
   - *Pentester Relevance:* This is where ARP Spoofing attacks happen.
3. **Layer 3 (Network):** Uses IP Addresses to route data between different networks.
   - *Pentester Relevance:* This is where we perform Host Discovery using tools like ping.
4. **Layer 4 (Transport):** Uses TCP/UDP to ensure data gets to the right application.
   - *Pentester Relevance:* This is the layer we target during Port Scanning.
5. **Layers 5-7 (Session, Presentation, Application):** Where the data is actually used (HTTP, FTP, SSH).
   - *Pentester Relevance:* This is where we perform Service Enumeration and Password Attacks.

1.3.6 TCP vs UDP

Computers communicate in two main ways. As a pentester, you need to know the difference because your scanning tools (like Nmap) behave differently for each.

TCP (Transmission Control Protocol)

Reliable: You care if the message arrives.
How it works (3-Way Handshake):

1. **SYN:** "Hey, are you there?"
2. **SYN-ACK:** "Yes, I am here."
3. **ACK:** "Great, let's talk."

This is called the 3-Way Handshake. Most scans look for this "handshake" to find open ports.

UDP (User Datagram Protocol)

Unreliable: You send the data and hope it gets there. There is no confirmation.

Streaming video often uses UDP. It is harder to scan because the server doesn't always reply to you.

1.3.7 Ports (Where Services Run)

If an IP address is the "House," a Port is a "Door" or "Window" into that house. Each service (like a website or file share) lives behind a specific door number.

Our Goal: Find the doors that are open.

Example

If a server has:

- Port 80 open → Website running (HTTP)
- Port 22 open → Remote access available (SSH)
- Port 445 open → File sharing service (SMB)

Important Note

Ports do not always run their default services. Administrators can reconfigure any service to use a different port, so enumeration and service-version detection are essential.

These open doors tell us what services the machine is running and what attack paths may exist.

1.3.8 DNS (Domain Name System)

Computers don't speak English, they speak numbers (IP Addresses). Humans, however, are bad at remembering numbers like `142.250.190.46`, but we are good at remembering names like `google.com`.

DNS (Domain Name System) translates the human name into the computer number.

How it Works (The Contacts List)

Imagine you want to call your friend "Alice." You don't memorize her number; you just tap "Alice" in your contacts list.

1. **You:** Type `www.facebook.com`
2. **DNS Server:** Looks up "Facebook" in its massive list
3. **Reply:** "Facebook lives at `157.240.221.35`"
4. **Result:** Your computer connects to that IP address

1.4 Linux Command Line Essentials

As a penetration tester, you'll spend most of your time in Linux terminals (particularly Kali Linux). Mastering basic Linux commands is not optional — it's essential. This section covers the commands you'll use daily during the eJPT exam.

1.4.1 Why Linux for Penetration Testing?

- Most penetration testing tools are built for Linux
- The eJPT exam provides a Kali Linux environment
- Better control and automation capabilities
- Open-source with powerful command-line tools

1.4.2 Linux File System Structure

Understanding where things live in Linux is crucial:

Important Directories

- `/home/[username]` — Your personal files and workspace
- `/root` — Root user's home directory (administrator)
- `/etc` — Configuration files for the system
- `/tmp` — Temporary files (often writable by everyone)
- `/var` — Variable data (logs, databases)
- `/usr/bin` — User programs and applications
- `/opt` — Optional software packages

1.4.3 Navigation Commands

pwd (Print Working Directory)

Shows you where you currently are in the file system.

Example

```
$ pwd
/home/kali
```

ls (List)

Shows files and folders in your current location.

Common Usage

`$ ls` — Basic listing
`$ ls -l` — Long format (detailed)
`$ ls -la` — Include hidden files (files starting with .)
`$ ls -lh` — Human-readable file sizes

cd (Change Directory)

Move between folders.

Examples

`$ cd /home/kali` — Go to specific directory
`$ cd ..` — Go up one level
`$ cd ~` — Go to your home directory
`$ cd -` — Go back to previous directory

1.4.4 File and Directory Management

mkdir (Make Directory)

Create a new folder.

Example

```
$ mkdir pentest
$ mkdir -p notes/recon/nmap — Create nested directories
```

touch

Create an empty file or update timestamp.

Example

```
$ touch notes.txt
```

cp (Copy)

Copy files or directories.

Examples

`$ cp file1.txt file2.txt` — Copy file
`$ cp -r folder1 folder2` — Copy directory recursively

mv (Move/Rename)

Move or rename files.

Examples

`$ mv oldname.txt newname.txt` — Rename
`$ mv file.txt /tmp/` — Move to different location

rm (Remove)

Delete files or directories. **Be careful — there's no undo!**

Examples

`$ rm file.txt` — Delete file
`$ rm -r folder/` — Delete directory recursively
`$ rm -rf folder/` — Force delete (use with extreme caution!)

1.4.5 Viewing File Contents

cat (Concatenate)

Display entire file contents.

Example

```
$ cat passwords.txt
```

less

View file one page at a time (better for large files).

Example

```
$ less scan_results.txt
```

Navigation:
Space — Next page
b — Previous page
q — Quit

head and tail

View first or last lines of a file.

Examples

`$ head -n 10 file.txt` — First 10 lines
`$ tail -n 20 file.txt` — Last 20 lines
`$ tail -f /var/log/auth.log` — Follow file in real-time (useful for logs)

1.4.6 File Permissions

Linux uses a permission system to control who can read, write, or execute files.

Understanding Permissions

When you run `ls -l`, you see:

Example

```
-rwxr-xr-x 1 kali kali 4096 Dec 10 10:30 script.sh
```

Breaking down `-rwxr-xr-x`:

- `-` — File type (- = file, d = directory)
- `rwx` — Owner permissions (read, write, execute)
- `r-x` — Group permissions (read, execute)
- `r-x` — Others permissions (read, execute)

chmod (Change Mode)

Change file permissions.

Examples

`$ chmod +x script.sh` — Make file executable
`$ chmod 755 script.sh` — rwxr-xr-x
`$ chmod 644 file.txt` — rw-r–r–

Common Permission Numbers

- `777` — rwxrwxrwx (everyone can do everything)
- `755` — rwxr-xr-x (owner full, others read/execute)
- `644` — rw-r–r– (owner read/write, others read only)
- `600` — rw——- (only owner can read/write)

1.4.7 Text Searching and Manipulation

grep (Global Regular Expression Print)

Search for patterns in files. This is one of the most powerful commands for pentesting.

Examples

`$ grep "admin" users.txt` — Find lines containing "admin"
`$ grep -i "password" file.txt` — Case-insensitive search
`$ grep -r "config" /etc/` — Search recursively in directory
`$ grep -v "comment" file.txt` — Show lines NOT containing "comment"

find

Locate files by name, size, permissions, etc.

Examples

`$ find / -name "password.txt" 2>/dev/null` — Find file by name
`$ find /home -type f -size +10M` — Files larger than 10MB
`$ find / -perm -4000 2>/dev/null` — Find SUID files (privilege escalation)

cut

Extract specific columns from text.

Example

`$ cat users.txt | cut -d: -f1` — Extract first field (delimiter: colon)

sort and uniq

Organize and filter data.

Examples

`$ sort users.txt` — Sort alphabetically
`$ sort -u users.txt` — Sort and remove duplicates
`$ cat data.txt | sort | uniq` — Remove duplicates

1.4.8 Input/Output Redirection

Redirect command output to files or chain commands together.

Redirection Operators

- `>` — Redirect output (overwrite file)
- `>>` — Redirect output (append to file)
- `<` — Redirect input from file
- `|` — Pipe output to another command
- `2>` — Redirect errors
- `2>/dev/null` — Discard errors

Examples

`$ nmap 192.168.1.0/24 > scan.txt` — Save output to file
`$ cat users.txt | grep admin` — Pipe to grep
`$ find / -name "*.conf" 2>/dev/null` — Hide error messages

1.4.9 System Information Commands

whoami

Show current username.

Example

```
$ whoami
kali
```

id

Show user ID and group memberships.

Example

```
$ id
uid=1000(kali) gid=1000(kali) groups=1000(kali),27(sudo)
```

uname

Show system information.

Example

`$ uname -a` — Show all system info

```
Linux kali 6.1.0-kali5-amd64 #1 SMP Debian 6.1.12-1kali2 x86_64
```

ifconfig / ip

Show network interfaces.

Examples

`$ ifconfig` — Show network interfaces (older command)
`$ ip a` — Show network interfaces (modern command)
`$ ip route` — Show routing table

1.4.10 Network Commands

ping

Test connectivity to a host.

Example

`$ ping -c 4 192.168.1.1` — Send 4 packets
`$ ping 192.168.1.1` — Continuous ping (Ctrl+C to stop)

netstat

Show network connections.

Examples

`$ netstat -tuln` — Show listening ports
`$ netstat -ano` — Show all connections with PIDs

wget and curl

Download files from the web.

Examples

`$ wget http://example.com/file.txt` — Download file
`$ curl http://example.com` — Fetch webpage content
`$ curl -I http://example.com` — Get HTTP headers only

1.4.11 Process Management

ps (Process Status)

View running processes.

Examples

`$ ps aux` — Show all processes
`$ ps aux | grep apache` — Find specific process

kill

Terminate processes.

Examples

`$ kill 1234` — Kill process with PID 1234
`$ kill -9 1234` — Force kill
`$ killall firefox` — Kill all processes by name

1.4.12 Sudo (Superuser Do)

Run commands with administrator privileges.

Example

`$ sudo nmap -sS 192.168.1.1` — Run Nmap with root privileges
`$ sudo su` — Switch to root user
`$ sudo -l` — List sudo permissions (useful for privilege escalation)

eJPT Tip

During the exam, check what commands you can run with sudo using `sudo -l`. This is critical for privilege escalation!

1.4.13 Useful Shortcuts

Keyboard Shortcuts

- `Ctrl + C` — Stop current command
- `Ctrl + L` — Clear screen (same as `clear`)
- `Ctrl + R` — Search command history
- `Tab` — Auto-complete file/directory names
- `Up/Down arrows` — Navigate command history
- `Ctrl + A` — Jump to beginning of line
- `Ctrl + E` — Jump to end of line

1.5 Windows Command Line Essentials

While most penetration testing tools run on Linux, you'll often attack Windows systems. Understanding Windows command-line basics helps you navigate compromised systems, execute commands through shells, and perform post-exploitation tasks.

1.5.1 Two Windows Shells

Windows has two command-line interfaces:

1. **Command Prompt (cmd.exe)** — Traditional DOS-style shell
2. **PowerShell** — Modern, powerful scripting environment

eJPT Focus

For eJPT, focus primarily on **Command Prompt** basics. PowerShell is less commonly needed but good to know.

1.5.2 Opening Command Prompt

- Press `Win + R`, type `cmd`, press Enter
- Search for "cmd" or "Command Prompt" in Start Menu
- For admin access: Right-click → "Run as administrator"

1.5.3 Navigation Commands

cd (Change Directory)

Move between folders.

Examples

`C:\> cd Documents` — Enter folder
`C:\> cd ..` — Go up one level
`C:\> cd \` — Go to root directory
`C:\> cd /d D:\` — Change to different drive

dir

List files and folders (equivalent to Linux `ls`).

Examples

`C:\> dir` — List current directory
`C:\> dir /a` — Show hidden files
`C:\> dir /s` — Recursive listing (all subdirectories)

1.5.4 File and Directory Management

mkdir / md

Create new directory.

Example

```
C:\> mkdir pentest
```

copy

Copy files.

Examples

`C:\> copy file1.txt file2.txt` — Copy file
`C:\> copy *.txt D:\backup\` — Copy all .txt files

xcopy

Copy directories and contents.

Example

`C:\> xcopy /E /I source dest` — Copy directory recursively

move

Move or rename files.

Examples

`C:\> move old.txt new.txt` — Rename
`C:\> move file.txt C:\temp\` — Move file

del / erase

Delete files.

Examples

`C:\> del file.txt` — Delete file
`C:\> del /F /Q *.tmp` — Force delete all .tmp files

rmdir / rd

Remove directory.

Examples

`C:\> rmdir folder` — Remove empty folder
`C:\> rmdir /S /Q folder` — Remove folder and contents (force)

1.5.5 Viewing File Contents

type

Display file contents (equivalent to Linux `cat`).

Example

```
C:\> type passwords.txt
```

more

View file page by page.

Example

```
C:\> more logfile.txt
```

Space — Next page
Q — Quit

1.5.6 System Information Commands

hostname

Display computer name.

Example

```
C:\> hostname
Security-Eagle
```

whoami

Show current username.

Example

```
C:\> whoami
Security-Eagle\john
```

systeminfo

Display detailed system information.

Example

```
C:\> systeminfo
```

This shows OS version, patches, hardware specs — critical for finding exploits!

wmic

Windows Management Instrumentation Command (powerful information gathering).

Examples

`C:\> wmic qfe list` — List installed patches
`C:\> wmic product get name` — List installed software
`C:\> wmic os get caption` — Get OS version

1.5.7 Network Commands

ipconfig

Show network configuration (equivalent to Linux `ifconfig`).

Examples

`C:\> ipconfig` — Show IP addresses
`C:\> ipconfig /all` — Detailed network info
`C:\> ipconfig /displaydns` — Show DNS cache

ping

Test connectivity.

Example

`C:\> ping 192.168.1.1`
`C:\> ping -n 10 192.168.1.1` — Send 10 packets

netstat

Show network connections.

Examples

`C:\> netstat -ano` — Show all connections with PIDs
`C:\> netstat -an` — Show listening ports

arp

View ARP cache.

Examples

`C:\> arp -a` — Display ARP table

nslookup

DNS lookup tool.

Example

```
C:\> nslookup google.com
```

tracert

Trace route to destination (equivalent to Linux `traceroute`).

Example

```
C:\> tracert 8.8.8.8
```

1.5.8 User and Group Management

net user

Manage user accounts.

Examples

`C:\> net user` — List all users
`C:\> net user john` — Show user details
`C:\> net user john password123 /add` — Create new user
`C:\> net user john /delete` — Delete user

net localgroup

Manage groups.

Examples

`C:\> net localgroup` — List all groups
`C:\> net localgroup Administrators` — Show admin group members
`C:\> net localgroup Administrators john /add` — Add user to admin group

1.5.9 Service and Process Management

tasklist

List running processes (equivalent to Linux `ps`).

Examples

`C:\> tasklist` — Show all processes
`C:\> tasklist | findstr "chrome"` — Find specific process

taskkill

Terminate processes.

Examples

`C:\> taskkill /PID 1234` — Kill process by PID
`C:\> taskkill /F /IM chrome.exe` — Force kill by name

sc (Service Control)

Manage Windows services.

Examples

`C:\> sc query` — List all services
`C:\> sc query wuauserv` — Query specific service
`C:\> sc stop wuauserv` — Stop service
`C:\> sc start wuauserv` — Start service

1.5.10 File Searching

where

Find files by name.

Example

`C:\> where /R C:\password.txt` — Search recursively

findstr

Search text within files (equivalent to Linux `grep`).

Examples

`C:\> findstr "password" file.txt` — Find pattern in file
`C:\> findstr /S /I "admin" *.txt` — Search all .txt files (case-insensitive)

1.5.11 Output Redirection

Similar to Linux:

Redirection Operators

- `>` — Redirect output (overwrite)
- `»` — Redirect output (append)
- `|` — Pipe output to another command

Examples

`C:\> systeminfo > info.txt` — Save output to file
`C:\> dir | findstr ".txt"` — Pipe to findstr
`C:\> tasklist » processes.txt` — Append to file

1.5.12 PowerShell Basics (Bonus)

While cmd.exe is sufficient for eJPT, knowing a few PowerShell commands is helpful.

Launching PowerShell

- Type `powershell` in cmd.exe
- Press `Win + R`, type `powershell`

Common PowerShell Commands

Examples

`PS> Get-ChildItem` — List files (like dir)
`PS> Get-Process` — List processes
`PS> Get-Service` — List services
`PS> Get-NetIPAddress` — Show IP addresses
`PS> Test-Connection 192.168.1.1` — Ping host

1.5.13 Privilege Escalation Checks

When you compromise a Windows system, check these immediately:

Quick Wins for Privilege Escalation

1. `whoami /priv` — Check your privileges
2. `whoami /groups` — Check group memberships
3. `net user` — List all users
4. `net localgroup administrators` — Who has admin access?
5. `systeminfo` — Check patch level
6. `wmic qfe list` — List installed patches
7. `sc query` — Check running services

1.5.14 Useful Tips

Windows CMD Tips

- Windows paths use backslashes: `C:\Users\`
- Commands are case-insensitive: `DIR` = `dir`
- Use `Tab` for auto-completion
- Use `Up/Down arrows` for command history
- `cls` clears the screen
- `exit` closes the command prompt
- Use quotes for paths with spaces: `cd "Program Files"`

1.5.15 Common Mistakes to Avoid

- Forgetting `/d` when changing drives with cd: use `cd /d D:\`
- Using forward slashes instead of backslashes in paths
- Not using quotes around paths with spaces
- Forgetting that `del` is permanent (no recycle bin from cmd!)

eJPT Exam Reminder

During the exam, you'll primarily use Linux (Kali) to attack Windows targets. You'll use Windows commands when you get a shell on a compromised Windows system. Focus on information gathering commands like `systeminfo`, `whoami`, and `net user`.

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